

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 7

Subject Code & Name: BP704T - NOVEL DRUG DELIVERY SYSTEM

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

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|---|---|
| A) Enlist any four ideal properties of bioadhesive polymers. | 2 |
| B) Enlist the types of ocuserts. | 2 |
| C) What is mucoadhesion? | 2 |
| D) Write any four applications of microencapsulation. | 2 |
| E) Classify liposomes according to structure. | 2 |
| F) What are the disadvantages of conventional ocular drug delivery systems? | 2 |
| G) Differentiate between active targeting and passive targeting. | 2 |
| H) Enlist factors affecting in the designing of modified drug delivery systems. | 2 |
| I) State any two factors affecting drug permeation through the skin. | 2 |
| J) Define pulmonary drug delivery system. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

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| A) Explain in detail the different methods for formulation of transdermal drug delivery systems along with their evaluation. | 10 |
| B) Explain in detail the formulation methods and evaluation of nanoparticles along with the advantages of nanoparticulate drug delivery. | 10 |
| C) Describe various theories of mucoadhesion with their significance in designing mucoadhesive drug delivery systems. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

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| A) What are ion-exchange resins? Explain their mechanism. | 5 |
| B) Describe vapour pressure-activated implantable drug delivery devices. | 5 |
| C) Explain metered dose inhaler (MDI). | 5 |
| D) Explain intrauterine drug delivery systems with examples, and state their advantages and applications. | 5 |
| E) Explain the preparation and applications of monoclonal antibodies. | 5 |
| F) Explain glass transition temperature and thermogravimetric analysis (TGA) of polymers. | 5 |
| G) Define microencapsulation and explain any two techniques of microencapsulation. | 5 |
| H) Explain the approaches for controlled release formulations based on dissolution. | 5 |
| I) Define controlled drug delivery system. Explain the rationale, advantages, and disadvantages of controlled release drug delivery systems. | 5 |

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